

# Yadnesh Krishnakant Bhanushali

Applied AI / ML Engineer | AWS Architecture | Generative AI | Search Systems

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## Professional Summary

Applied AI / Machine Learning Engineer experienced in designing and deploying production AI systems across computer vision, generative AI, hybrid search, and cloud-based industrial platforms. Skilled at translating ambiguous business requirements into scalable architectures and delivering end-to-end solutions spanning models, data pipelines, cloud infrastructure, and monitoring systems. Known for strong ownership, rapid execution, and independent delivery of enterprise AI solutions in real client environments.

## Technical Skills

**Cloud & Architecture:** AWS (SageMaker, Bedrock, RDS, VPC, IAM, S3), Cloud Architecture Design, Infrastructure Security

**AI / ML:** LLMs, VLMs, Computer Vision, Generative AI, Real-ESRGAN, Learning-to-Rank (LTR), Hybrid Search, Data Modelling, Data Pre-Processing

**Search & Data Systems:** OpenSearch, BM25, Semantic kNN Search, Re-ranking Pipelines, Data Engineering

**Programming:** Python, Java, SQL, C++, PostgreSQL, HTML, CSS **Tools:** Grafana, Streamlit, Django, Git

## Experience

### Oneture Technologies

Oct 2025 – Present

*Associate Machine Learning Engineer (Converted from ML Intern)*

*Mumbai, India*

- **Hybrid AI Search Engine (PoC):** Sole developer of a high-throughput OpenSearch hybrid retrieval system on AWS, combining BM25 lexical search with semantic kNN embeddings, custom neural re-ranking, and Learning-to-Rank personalization — designed to handle **large-scale retrieval workloads** across multiple e-commerce user personas.
- **Multimodal AI Pipeline — Snapdeal & Nykaa (PoC → Pilot → Live):** Built and took to production a Vision Language Model system via AWS Bedrock that extracts structured product attributes, banner elements, semantic callouts, layout compliance scores, and visual metadata from large-scale e-commerce imagery. System is fully config-driven — operators control target detection zones, supported aspect ratios, and extraction rules from a single config file. Outputs downloadable Excel reports with extracted attributes, compliance flags, and layout scores. **Significantly reduced manual cataloguing effort by 80 %**, with zero human intervention required in the live pipeline after a 3-month pilot.
- **Generative Vision Pipelines:** Built SageMaker ML pipelines for image super-resolution and outpainting using Real-ESRGAN, enabling automated marketing image transformation at scale; deployed with a Streamlit GUI for non-technical stakeholder use.
- **Digital Twin Platform — Wind Energy (Live):** Contributed to AWS-based architecture for an industrial wind turbine Digital Twin platform covering telemetry ingestion, condition monitoring, and predictive analytics readiness. Developed multi-level Grafana dashboards (site, turbine, component) used by real site operators in production.
- **TEAL Demand Forecasting Platform:** Sole owner of database design and deployment for a client demand forecasting system. Planned, designed, and deployed AWS RDS per client specifications across secure VPC/subnet configurations. Built three ingestion pipelines: (1) bulk upload for initial data load, (2) record update pipeline for corrections and amendments, and (3) an incremental upsert pipeline for monthly data — intelligently updating quantity for existing inventory and sales records rather than overwriting, and inserting only new entries, preventing data duplication and preserving historical integrity.
- **Mentorship:** Guided two junior interns on ML implementation tasks alongside delivery of production projects.

## Projects

**AyurGenix AI** 2024–2025 🐙 GitHub — AI-powered Ayurvedic advisory system using LLM reasoning and ensemble ML models. Full-stack pipeline with Django and PostgreSQL; achieved **88% treatment mapping accuracy**. Published at ICETCSE-2025.

**PDF MasterMind** 2024 🐙 GitHub — LLM-powered document assistant for conversational querying and summarization of large PDFs using LangChain and transformer-based retrieval pipelines.

## Education

### Datta Meghe College of Engineering

*B.E. Artificial Intelligence & Data Science (Honors in Machine Learning)*

2021 – 2025

*CGPA: 9.04*

### K.V. Pendharkar College of Science

*Higher Secondary Certification (PCMB)*

2019 – 2021

*85%*

## Achievements & Publications

**Research Publication:** Presented "AyurGenix AI — Ayurvedic Medication Advisor" at the **International Conference on Emerging Trends in Computer Science and Engineering (ICETCSE-2025)**, Navi Mumbai, India.

**Hackathon:** Optimized algorithmic pipeline at CodeRush @ LogiTHON, improving solution efficiency by **20%**.